

GraphRAG vs. Standard RAG

A Practical Example Document Showing Why Entity-Relationship Awareness Matters

This PDF contains a fictitious but realistic corporate scenario about **ACME Corporation**. The information about people, projects, budgets, and decisions is deliberately spread across multiple sections, mimicking how real enterprise documents are structured. A standard chunk-based RAG system will struggle to answer multi-hop questions about this content, while a GraphRAG approach (which extracts entities and relationships into a knowledge graph) can synthesize answers that require connecting facts from distant parts of the document.

Document Type: Internal Corporate Memo + Org Analysis

Company: ACME Corporation

Purpose: Demonstrate multi-hop reasoning failure in standard RAG

1. Organizational Structure - ACME Corporation

ACME Corporation is a mid-size enterprise software company headquartered in Munich, Germany, with satellite offices in Milan (Italy), Austin (Texas, USA), and Bangalore (India). The company was founded in 2011 by Dr. Elena Vasquez, who currently serves as CEO. ACME operates across four divisions: Cloud Platform, Data Analytics, Enterprise AI, and Security.

1.1 Executive Leadership

Dr. Elena Vasquez (CEO) reports directly to the Board of Directors. The C-suite includes: Marco Rinaldi (CTO), who joined ACME in 2015 after a decade at SAP; Dr. Priya Sharma (Chief Data Officer, CDO), who leads the Data Analytics and Enterprise AI divisions; and Thomas Weiss (CFO), a former McKinsey partner responsible for all financial operations. The Chief Security Officer is Kenji Tanaka, who oversees both the Security division and internal compliance. Tanaka previously worked at CrowdStrike and reports to the CTO on technical matters and to the CEO on governance matters (a dual-reporting structure).

1.2 Division Heads and Key Personnel

Division	Head	Location	Headcount
Cloud Platform	Sofia Andersen	Munich	112
Data Analytics	Raj Patel	Bangalore	87
Enterprise AI	Dr. Lena Hoffmann	Munich + Austin	64
Security	Kenji Tanaka (acting)	All offices	41

Raj Patel reports to Dr. Priya Sharma. Dr. Lena Hoffmann also reports to Sharma but has a dotted-line relationship with the CTO (Marco Rinaldi) for infrastructure decisions. Sofia Andersen reports directly to Rinaldi. The Milan office is led by Alessandro Bianchi, who manages a cross-functional team of 23 engineers working on both Cloud Platform and Data Analytics projects.

2. Active Projects - Q3/Q4 2025

2.1 Project Aurora (Cloud Platform)

Project Aurora is ACME's next-generation multi-cloud orchestration platform, led by Sofia Andersen with a budget of EUR 4.2 million for FY2025. The project began in January 2025 and has a target launch date of March 2026. Aurora's principal architect is Yuki Sato (Austin office), who designed the event-driven microservices backbone. The Milan sub-team, managed by Alessandro Bianchi, is responsible for the Kubernetes autoscaling module (codenamed 'Aurora-K8s'). In August 2025, Aurora-K8s experienced a critical regression that delayed the overall timeline by six weeks. Bianchi escalated the issue to Rinaldi, who authorized an additional EUR 300,000 contingency budget drawn from the Security division's unallocated Q4 funds.

2.2 Project Helios (Enterprise AI)

Project Helios is an LLM-powered document intelligence suite for enterprise customers. Dr. Lena Hoffmann leads the project with a team of 28 researchers and engineers split between Munich and Austin. The project budget is EUR 3.8 million, funded partly by an EU Horizon grant (EUR 1.1 million) secured by Dr. Priya Sharma in Q1 2025. The grant requires the project to use open-source models only, which has created a technical constraint: Dr. Hoffmann's team initially planned to fine-tune a proprietary model from Anthropic but pivoted to Mistral-based architectures after the grant condition was imposed. The Bangalore team (under Raj Patel) is contributing a data-labeling pipeline for Helios, creating a dependency between the Data Analytics and Enterprise AI divisions.

2.3 Project Sentinel (Security)

Project Sentinel is a zero-trust network access (ZTNA) product being developed under Kenji Tanaka's direction. The budget is EUR 1.9 million, but after Rinaldi redirected EUR 300,000 to Aurora (see Section 2.1), the effective Sentinel budget for Q4 is EUR 1.6 million. Tanaka has formally objected to this reallocation in a memo to CFO Thomas Weiss, arguing that the reduced budget jeopardizes Sentinel's SOC2 certification timeline. Weiss has proposed a compromise: restore EUR 150,000 from the company's general reserve if Sentinel meets its November milestone. The Sentinel team includes 12 engineers in Munich and 8 contractors in Bangalore, managed locally by Ananya Desai (who reports to both Tanaka and Raj Patel).

2.4 Project Nexus (Data Analytics)

Project Nexus is a real-time analytics engine being built by the Bangalore team. Raj Patel is the project lead, with Ananya Desai serving as technical lead (in addition to her Sentinel responsibilities). Nexus has a modest budget of EUR 1.2 million and is on track for a Q1 2026 release. However, 15% of the Bangalore engineering team's capacity is allocated to the Helios data-labeling pipeline (see Section 2.2), creating resource contention that Patel has flagged to Sharma in weekly standups. Nexus uses Apache Kafka and ClickHouse, technologies also adopted by the Aurora-K8s module - a technical overlap that Bianchi and Patel have discussed for potential code sharing but have not formalized.

3. Incident Report: Aurora-K8s Regression (INC-2025-0847)

On August 14, 2025, the Aurora-K8s autoscaling module deployed to the staging environment triggered an uncontrolled pod-scaling loop, consuming 340% of cluster resources within 18 minutes. The incident was detected by the monitoring stack maintained by the Security division's SRE sub-team (led by Ananya Desai, who also works on Sentinel and Nexus).

3.1 Root Cause

The root cause was a race condition in the horizontal pod autoscaler (HPA) configuration introduced by a commit from Milan-based engineer Giulia Ferraro. Ferraro had modified the cooldown period from 120 seconds to 15 seconds based on performance benchmarks conducted on a single-node test cluster that did not replicate the production topology. The change was approved in a code review by Yuki Sato (Austin), who later acknowledged insufficient attention to the multi-node implications.

3.2 Impact and Response

The staging outage lasted 47 minutes. No customer data was affected, but the incident exposed a gap in the cross-team code review process. Alessandro Bianchi (Milan lead) convened a post-mortem with Sato, Ferraro, and Desai. Key findings: (a) the Milan and Austin teams lacked a shared integration testing environment; (b) Desai's SRE monitoring detected the issue within 3 minutes but had no automated rollback capability for staging; (c) the six-week delay was primarily caused by the need to rebuild the HPA logic and set up a shared test environment. The cost of remediation - EUR 300,000 - was the amount Rinaldi pulled from Sentinel.

3.3 Follow-Up Actions

Rinaldi mandated that all cross-office commits require approval from at least one reviewer in each participating office. Hoffmann (Enterprise AI) offered to lend two MLOps engineers from her Helios team to help Bianchi's team set up a proper CI/CD pipeline for Aurora-K8s - a collaboration approved by Sharma since it also benefits the Helios deployment workflow. Tanaka added the Aurora-K8s incident as a case study in ACME's internal security training materials, noting the relevance to Sentinel's own multi-region deployment architecture.

4. Financial Summary - FY2025 Project Budgets

Project	Original Budget	Adjustments	Effective Q4	Funding Source
Aurora	EUR 4,200,000	+300,000	EUR 4,500,000	Internal + Sentinel realloc.
Helios	EUR 3,800,000	None	EUR 3,800,000	Internal + EU Horizon (1.1M)
Sentinel	EUR 1,900,000	-300K (+150K pending)	EUR 1,600,000	Internal
Nexus	EUR 1,200,000	None	EUR 1,200,000	Internal

Thomas Weiss (CFO) has noted in his Q3 board presentation that total project spend for FY2025 is EUR 11.1 million, representing 38% of ACME's total revenue. He flagged the Aurora cost overrun as a concern and recommended that any further budget reallocations require board-level approval rather than CTO discretion alone. Dr. Vasquez (CEO) supported this recommendation and instructed Weiss to draft a revised budget governance policy by November 2025.

4.1 EU Horizon Grant Compliance

The EUR 1.1 million EU Horizon grant that partially funds Helios is subject to strict compliance requirements: (a) all model components must be open-source; (b) at least 30% of the budget must be spent in EU member states; (c) quarterly progress reports must be submitted to the European Research Council. Dr. Sharma oversees grant compliance and has appointed Dr. Hoffmann as the technical compliance officer. In September 2025, an internal audit by Weiss's finance team discovered that 22% of Helios expenditure was allocated to the Austin office (a non-EU location), putting the 30% EU-spend requirement at risk. Sharma responded by shifting two Austin-based researchers to the Munich office for the remainder of the project.

5. Personnel Changes and HR Matters - Q3 2025

In July 2025, Sofia Andersen (Cloud Platform division head) announced her intention to take a six-month sabbatical beginning January 2026, immediately after Aurora's planned launch. Rinaldi has designated Yuki Sato as acting division head during Andersen's absence. However, Sato is based in Austin, which has raised concerns about timezone alignment with the Munich-heavy Cloud Platform team. Bianchi (Milan) has been proposed as an alternative interim lead by several Munich-based engineers, but Rinaldi prefers Sato due to his architectural expertise.

Ananya Desai (Bangalore) received a promotion to Senior Principal Engineer in August 2025, making her the youngest person at that level in ACME's history. However, her triple allocation across Sentinel, Nexus, and SRE monitoring has been flagged as a burnout risk by HR director Maria Schulz. Schulz recommended to Sharma and Tanaka that Desai's workload be reduced by Q1 2026. Tanaka agreed to reassign the SRE monitoring role to a new hire, pending budget approval from Weiss.

Giulia Ferraro (Milan), the engineer whose commit caused the Aurora-K8s incident, was placed on a performance improvement plan (PIP) by Bianchi in September 2025. Ferraro has disputed the PIP, arguing that Sato's code review approval shared responsibility for the regression. HR (Schulz) is mediating the dispute. Rinaldi has stated that the revised cross-office review policy (see Section 3.3) should prevent similar situations in the future.

6. Analysis: Why GraphRAG Outperforms Standard RAG on This Document

6.1 The Fundamental Problem with Chunk-Based RAG

Standard RAG systems split documents into fixed-size text chunks (e.g., 500 tokens) and retrieve the top-k most semantically similar chunks for a given query. This works well for factoid questions where the answer resides in a single chunk. However, it fails when answering a question requires synthesizing information from multiple distant parts of the document - so-called 'multi-hop' reasoning.

6.2 GraphRAG Approach

GraphRAG (as described in the Microsoft Research paper, 2024) adds an entity-extraction and relationship-mapping step before retrieval. An LLM reads the full document and builds a knowledge graph of entities (people, projects, locations, events) and their relationships. At query time, the system traverses this graph to gather all relevant context, even if the facts are spread across many sections. Community detection algorithms can also identify clusters of related entities, enabling high-level summarization queries that standard RAG cannot address.

6.3 Example Multi-Hop Questions

Below are five questions that require information from two or more sections. For each, we show what a Standard RAG chatbot would likely answer (limited to the best-matching chunk) vs. what a GraphRAG chatbot could answer (using entity traversal).

Q1: What is the total budget impact of the Aurora-K8s incident on other projects?

Standard RAG: "The Aurora-K8s incident caused a six-week delay and cost EUR 300,000 in remediation." (Retrieves Section 3 only - misses the impact on Sentinel and the CFO governance response.)

GraphRAG: "The EUR 300,000 remediation cost was funded by reallocating from Sentinel's Q4 budget (reducing it from EUR 1.9M to EUR 1.6M). This prompted Tanaka to formally object to CFO Weiss, who proposed a conditional EUR 150K restoration. The overrun also led the CEO to mandate board-level approval for future reallocations, changing ACME's budget governance policy."

Q2: Which person works across the most projects, and what risk does this create?

Standard RAG: "Ananya Desai works on multiple projects in Bangalore." (Retrieves one section but cannot connect all three roles or the HR concern.)

GraphRAG: "Ananya Desai is allocated across three areas: Sentinel (security), Nexus (data analytics), and SRE monitoring. She reports to both Tanaka and Patel. HR director Schulz has flagged her workload as a burnout risk, and Tanaka agreed to reassign her SRE role to a new hire, pending CFO budget approval."

Q3: How does the EU Horizon grant constrain technical decisions in Project Helios?

Standard RAG: "Project Helios uses Mistral-based architectures." (Retrieves part of Section 2.2, misses the grant compliance chain from Section 4.1.)

GraphRAG: "The EU grant (EUR 1.1M, secured by Sharma) requires open-source models, forcing Hoffmann's team to pivot from Anthropic proprietary models to Mistral. It also mandates 30% EU spend, which was jeopardized when an audit found 22% going to Austin, prompting Sharma to relocate two researchers to Munich."

Q4: If Sofia Andersen goes on sabbatical, what are the organizational risks for Aurora?

Standard RAG: "Yuki Sato will serve as acting division head during Andersen's sabbatical." (Retrieves Section 5 only, misses Sato's role in the incident and the timezone concern.)

GraphRAG: "Sato (Austin) is designated as acting head, but he was also the reviewer who approved the faulty Aurora-K8s commit. Munich engineers prefer Bianchi (Milan) instead. There is a timezone gap: Cloud Platform is Munich-heavy, but Sato is in Austin. Additionally, Bianchi already manages a cross-functional Milan team, so either choice creates capacity risks."

Q5: What are all the cross-division dependencies in ACME's project portfolio?

Standard RAG: "The Bangalore team contributes a data-labeling pipeline for Helios." (Returns one chunk, misses the full dependency web.)

GraphRAG: "There are at least five cross-division dependencies: (1) Bangalore Data Analytics provides the labeling pipeline for Helios (Enterprise AI); (2) Sentinel budget was raided for Aurora (Cloud Platform); (3) Desai (Security/SRE) also works on Nexus (Data Analytics); (4) Hoffmann (Enterprise AI) lent MLOps engineers to Aurora-K8s (Cloud Platform); (5) Nexus and Aurora-K8s share Kafka/ClickHouse technology."

7. Entity-Relationship Summary (What GraphRAG Extracts)

Below is a simplified view of the entities and relationships that a GraphRAG pipeline would extract from Sections 1-5 of this document. This is the knowledge graph that enables the multi-hop answers shown in Section 6.

7.1 Key Entities

Type	Entities
People	Dr. E. Vasquez (CEO), M. Rinaldi (CTO), Dr. P. Sharma (CDO), T. Weiss (CFO), K. Tanaka (CSO), S. Andersen, R. Patel, Dr. L. Hoffmann, A. Bianchi, Y. Sato, A. Desai, G. Ferraro, M. Schulz
Projects	Aurora (+ Aurora-K8s sub-module), Helios, Sentinel, Nexus
Locations	Munich (HQ), Milan, Austin, Bangalore
Organizations	ACME Corporation, EU Horizon Grant Program, European Research Council
Events	Aurora-K8s Regression (INC-2025-0847), Andersen Sabbatical, Ferraro PIP, Desai Promotion, Budget Reallocation Dispute

7.2 Critical Relationship Chains (Multi-Hop Paths)

#	Relationship Chain	Sections
1	Ferraro (commit) -> Aurora-K8s (regression) -> Rinaldi (budget transfer) -> Sentinel (reduced budget) -> Tanaka (objection) -> Weiss (compromise)	2.1, 3.1, 3.2, 2.3, 4
2	EU Grant (Sharma) -> Helios (open-source constraint) -> Hoffmann (pivot to Mistral) -> Austin spend audit -> researcher relocation to Munich	2.2, 4.1
3	Desai -> Sentinel + Nexus + SRE -> burnout risk (Schulz) -> Tanaka agrees to reassign SRE -> pending Weiss budget approval	2.3, 2.4, 3, 5
4	Andersen (sabbatical) -> Sato (acting head, Austin) -> timezone concern -> Bianchi (alternative) -> Rinaldi (prefers Sato)	1.2, 2.1, 5

7.3 Key Takeaway

Standard RAG retrieves text fragments. GraphRAG retrieves entity subgraphs. When a question's answer requires traversing a chain of relationships (person -> project -> event -> budget -> policy), GraphRAG can follow that chain through the knowledge graph, while standard RAG is limited to whatever facts happen to appear in the highest-scoring chunk. For documents like this one, where the narrative is structured around interconnected organizational entities, GraphRAG consistently produces more complete, accurate, and contextually rich answers.

Document generated for demonstration purposes. All names, companies, and events are fictitious.